Mitchell Tillman, PhD

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Objective

Postdoctoral fellow seeking a Program Manager clinical biomechanist role, where patient-facing assessment and data analysis go hand in hand. My postdoc has combined delivering tSCS in the Tech Immersion clinic with device selection, rollout monitoring, and outcomes analysis for its hospital-wide implementation, building on Ph.D. training in whole-body gait biomechanics. My vision is for data to empower movement.

Experience

- **Shirley Ryan AbilityLab** *2024 – Present*
  Postdoctoral Research Fellow, Center for Bionic Medicine Chicago, IL USA
  - Study the effects of transcutaneous spinal cord stimulation (tSCS) on gait in people with stroke and traumatic brain injury in Dr. Arun Jayaraman's lab
  - Provide technical expertise to the Tech Immersion clinic, collecting and analyzing sensor data (EMG, IMU) to inform care for medically complex patients, and delivering tSCS treatment to spinal cord injury patients
  - Assist the team implementing tSCS into hospital-wide standard clinical practice for spinal cord injury patients by informing stimulator device selection, designing clinician education, monitoring rollout operations, and analyzing clinical tSCS outcomes
  - Co-authored three journal articles on tSCS in chronic stroke in 2026, including a first-author article in *Neuromodulation*
- **Stevens Institute of Technology** *2019 – 2024*
  Graduate Research Assistant, MSKCD Lab Hoboken, NJ USA
  - Dissertation research with Dr. Antonia Zaferiou quantifying balance and momentum control during 90° turns while walking in healthy young and older adults. Quantified several types of balance maintenance strategies
  - Collected motion capture and force plate data on over 50 young and older adult participants walking and turning
  - Developed a MATLAB data analysis pipeline for custom analyses based on person-specific biomechanical models, including bespoke visualizations and data-driven videos. Designed the pipeline to handle large data volumes by minimizing the data held in memory
  - Led three first-author journal articles from experiment design through data analysis and manuscript writing
- **Purdue University** *2016 – 2018*
  Research Assistant, Biomechanics Lab West Lafayette, IN USA
  - Three-first author publications detailing how anticipation affects stability in finger force-production tasks under the guidance of Dr. Satyajit Ambike, collecting force transducer data and analyzing movement variability

Projects

- **SciStack: Scientific data processing infrastructure (public beta)** *Sep 2021 – Present*
  Tools: Python, MATLAB, SQL (DuckDB), VS Code extension
  - Developed open-source software infrastructure to power researchers' data analyses, so researchers can focus more on their science, and less on the code around the science
  - Provides Python and MATLAB tools to streamline research workflows. A VS Code extension GUI supports interactive processing, plotting, and statistical output generation, with interactive debugging
  - Stores an auditable trail of all generated research outputs, facilitating reproducible science
  - This software provides tools that serve as a model for how a software framework could facilitate code sharing, increasing the transparency, reproducibility, and quality of research outcomes
- **Real-time auditory biofeedback to assist balance** *2019 – 2022*
  Tools: MATLAB, OptiTrack motion capture, MaxMSP, Open Sound Control; Python, DeepLabCut Live
  - **Markered motion capture:** Developed the first prototype of a real-time marker-based motion capture auditory biofeedback system (~20 ms latency). 3 of 5 young adults improved one-legged standing balance in pilot testing
  - **Markerless motion capture:** Led a team of 10 high school students (NJ Governor's STEM Scholars, 2021 – 2022) in developing a low-cost version using free software, mentored by Dr. Gary Kane, a core developer of DeepLabCut Live

Education

- **Stevens Institute of Technology** — *Oct 2024*
  Ph.D. Mechanics of Whole-Body Balance and Momentum Control During Straight-Line Gait and 90° Turns — Hoboken, NJ USA — GPA: 4.00/4.00
- **Purdue University** — *May 2019*
  M.S. Effects of Past and Future Motor Events on Present Motor Stability, and Relationships with Motor and Cognitive Flexibility — West Lafayette, IN USA — GPA: 3.59/4.00
- **Purdue University** — *May 2018*
  B.S. Movement & Sports Sciences — West Lafayette, IN USA — GPA: 3.12/4.00

Selected Publications

[1] **Tillman M**, et al. (2026). **Acute effects of transcutaneous spinal cord stimulation (tSCS) on gait symmetry in people with chronic stroke**. *Neuromodulation*

[2] Kishta A, Khazanchi R, Veit N, **Tillman M**, et al. (2026). **Multimodal machine learning to personalize transcutaneous spinal cord stimulation for stroke rehabilitation**. *Digital Biomarkers*

[3] Veit NC, et al., **Tillman M**, et al. (2026). **The effects of electrode configuration and location on motor thresholds, tolerance, and muscle selectivity of transcutaneous spinal cord stimulation in stroke**. *J. Neurophysiol.*

[4] **Tillman M**, et al. (2024). **Healthy older adults generate transverse-plane momenta required for 90 degree turns during the same phases of gait as in straight-line gait**. *J. NeuroEng. Rehabil.*, 21:145

Full publication list: mitchelltillman.com/publications

Skills

- **Programming Languages (in order of experience):** MATLAB, Python, R, SQL, bash
- **Databases & Version Control:** SQLite, DuckDB, Git, GitHub
- **Statistics:** Regression, ANOVA, k-means cluster analysis, linear mixed models
- **Lab & Clinical Technology:** OptiTrack motion capture, force plates, transcutaneous spinal cord stimulation (tSCS), electromyography (Delsys), inertial measurement units (XSENS), DeepLabCut (markerless motion capture), DeepLabCut Live, MaxMSP
- **Research, Clinical & Leadership:** Experiment design, research study management, team management, data collection, working with patients, clinical data analysis, clinically-relevant analyses, pipeline development, biomechanics

Honors and Awards

- **Doctoral Excellence Fellowship** — *May 2023*
  Stevens Institute of Technology – Full tuition and stipend for one academic year
- **Outstanding Masters Student in the College of Health and Human Sciences** — *May 2019*
  Purdue University – $500 for outstanding achievements in original research

Volunteer Experience

- **Student Committee Member** — *2022 – 2024*
  American Society of Biomechanics (ASB)
  - Helped plan student sessions at the international ASB conference and secured $5,000 sponsorship from OptiTrack two years running for the student night out
- **Secretary, Graduate Biomedical Engineering Society** — *Aug 2021 – Aug 2022*
  Stevens Institute of Technology
  - Instituted biweekly graduate student-led department seminars that are still ongoing
  - Hosted career development events for the BME department, including negotiating funding from the school for food and external speakers

References

Dr. Antonia Zaferiou *(Ph.D. Supervisor)*
Assistant Professor, Biomedical Engineering, Stevens
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Dr. Sean Sanford *(Former Colleague)*
Senior Verification Test Engineer, Johnson & Johnson MedTech
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Dr. Satyajit Ambike *(M.S. Supervisor)*
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